GCSE Mathematics - Foundation Tier

Paper 3 (Calculator) Practice - Set 8

Time: 1 hour 30 minutes | Total marks: 80 | Calculator allowed

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Instructions

- Use black ink or ball-point pen.
- Answer ALL questions in the spaces provided.
- You must show all your working - answers given without working may not score full marks.
- Diagrams are NOT accurately drawn unless otherwise stated.
- If your calculator does not have a pi button, use pi = 3.142.

1.

Work out 10% of £85.

(Total for Question 1 is 1 marks)

2.

Write the improper fraction 23/4 as a mixed number.

(Total for Question 2 is 2 marks)

3.

(a) Round 4837 to 3 significant figures. (1)

(b) Round 348 to the nearest 100. (1)

(Total for Question 3 is 2 marks)

4.

Write the ratio 200 g : 1.5 kg in its simplest form. (Give the ratio in the form a : b where a and b are whole numbers.)

(Total for Question 4 is 2 marks)

5.

A library has 240 books, divided between fiction and non-fiction. The two-way table shows some of the data.

	Fiction	Non-fiction	Total
Adult	85	___	150
Children	___	40	___
Total	___	___	240

(a) Complete the two-way table. (2)

(b) A book is chosen at random. Work out the probability that the book is an adult non-fiction book. (1)

(Total for Question 5 is 3 marks)

6.

Solve $x - 4 = -7$.

(Total for Question 6 is 2 marks)

7.

A TV has a screen with a diagonal of 50 cm and a width of 40 cm.

[Diagram: rectangle representing the TV screen: width 40 cm; diagonal 50 cm; height labelled h cm.]

Work out the height of the screen.

(Total for Question 7 is 3 marks)

8.

The number of cars parked in a small car park at midday over 9 days are recorded.

14 12 18 17 12 20 15 19 12

(a) Find the mode. (1)

(b) Find the range. (1)

(c) Find the mean. Give your answer to 1 decimal place. (1)

(Total for Question 8 is 3 marks)

9.

A ship sails from port P on a bearing of 110 deg.

(a) Express the bearing 110 deg as a compass description (giving direction relative to S or N). (1)

A second port Q is on a bearing of 290 deg from port P.

(b) Work out the bearing of P from Q. (2)

(Total for Question 9 is 3 marks)

10.

The formula $y = 2x^2 + 5$ is given.

(a) Work out y when $x = -3$. (2)

The formula $y = (a + b) / 2$ is also given.

(b) Work out y when $a = 8$ and $b = -12$. (1)

(Total for Question 10 is 3 marks)

11.

A cube has a volume of 125 cm^3.

(a) Work out the length of one side of the cube. (2)

(b) Work out the total surface area of the cube. (1)

(Total for Question 11 is 3 marks)

12.

Yusuf invests £6500 in a savings account. The account pays compound interest at 2% per year.

Work out the total value of the investment after 3 years. Give your answer to the nearest penny.

(Total for Question 12 is 4 marks)

13.

A spinner has four equal sections numbered 1, 2, 3 and 4. The spinner is spun twice.

(a) Write down the total number of possible outcomes. (1)

(b) Work out the probability that the total of the two spins is exactly 5. (3)

(Total for Question 13 is 4 marks)

14.

The cross-section of a roof is an isosceles triangle. The base of the triangle is 6 m wide. Each of the two equal sloping sides (rafters) is 5 m long.

[Diagram: isosceles triangle with base 6 m and two equal sides of 5 m; vertical height h from apex to centre of base.]

(a) Show that the perpendicular height h is 4 m. (3)

(b) Hence work out the area of the cross-section. (2)

(Total for Question 14 is 5 marks)

15.

A box in the shape of a closed cuboid has dimensions 30 cm by 20 cm by 15 cm.

(a) Work out the volume of the box. (2)

(b) Work out the total surface area of the box. (3)

(Total for Question 15 is 5 marks)

16.

The nth term of a sequence is given by $T_n = 6n - 1$.

(a) Write down the 5th term. (1)

(b) Work out the sum of the 1st and 10th terms. (2)

(c) Find which term equals 89. (1)

(Total for Question 16 is 4 marks)

17.

A factory tested 8 batches of light bulbs. The table shows the average usage time (hours) versus the percentage of bulbs that failed within 100 hours.

Usage	400	450	500	550	600	650	700	750	
Failed	12	10	9	7	6	5	3	2	(%)

(a) On the grid provided, plot these data as a scatter graph. (2)

(b) Describe the correlation. (1)

(c) Draw a line of best fit. (1)

(d) Use your line of best fit to estimate the failure rate for a usage time of 580 hours. (1)

(Total for Question 17 is 5 marks)

18.

A cylindrical tank has a radius of 0.5 m and a height of 2 m.

[Diagram: cylindrical tank of radius 0.5 m and height 2 m.]

(a) Work out the volume of the tank in cubic metres. Use pi = 3.142. Give your answer to 3 decimal places. (3)

The tank is filled with water. 1 m^3 of water has a mass of 1000 kg.

(b) Work out the mass of water in the tank when it is full. Give your answer in kilograms to 1 decimal place. (2)

(Total for Question 18 is 5 marks)

19.

A coastguard station at point S records the position of a boat. The boat is 8 km due east of S, then sails 6 km due north.

[Diagram: right-angled triangle: S to A (east) 8 km, A to B (north) 6 km, distance S to B as hypotenuse.]

(a) Work out the straight-line distance from the boat to the coastguard station. (3)

(b) On a coastal map of scale 1 : 25 000, what would the straight-line distance from the boat to the station be (in cm)? (2)

(Total for Question 19 is 5 marks)

20.

A bag contains three numbered cards: 3, 4 and 7. A card is taken, the number recorded, and the card replaced. A second card is then taken.

Let S be the sum of the two numbers drawn.

(a) List all 9 possible (ordered) outcomes. (2)

(b) Show that the probability that S is even is exactly 5/9. (3)

(Total for Question 20 is 5 marks)

21.

Lola opens a savings account with £2000.

The account pays 5% compound interest per year.

At the end of each year, AFTER the interest has been added, Lola withdraws £150.

(a) Work out the balance at the end of year 1. (3)

(b) Work out the balance at the end of year 2. Give your answer to the nearest penny. (3)

(Total for Question 21 is 6 marks)

22.

A class of 32 students has 14 boys and 18 girls. The teacher chooses one student at random to read a passage.

(a) Write down the probability that the chosen student is a girl. Give your answer as a fraction in its simplest form. (2)

After the first student has read, the teacher chooses a second student at random. The first student is still in the class and could be chosen again.

(b) Work out the probability that the FIRST chosen student is a boy AND the SECOND is a girl. (3)

(Total for Question 22 is 5 marks)

TOTAL FOR PAPER: 80 MARKS

Mark Scheme

Foundation Tier - Paper 3 (Calculator) - Set 8 | 80 marks

Question 1

- £8.50 - B1

Total: 1 marks

Question 2

- 23 / 4 = 5 remainder 3 - M1
- 5 3/4 - A1

Total: 2 marks

Question 3

- (a) 4840 - B1
- (b) 300 - B1

Total: 2 marks

Question 4

- Convert 1.5 kg to 1500 g (or 200 g to 0.2 kg) and write 200 : 1500 - M1
- 2 : 15 - A1

Total: 2 marks

Question 5

- (a) Adult Non-fiction = 150 - 85 = 65 - M1
- (a) Children total = 240 - 150 = 90; Children Fiction = 90 - 40 = 50; column totals: Fiction 135, Non-fiction 105 - A1
- (b) 65 / 240 = 13/48 - B1

Total: 3 marks

Question 6

- Either +4 added to both sides - M1
- x = -3 - A1

Total: 2 marks

Question 7

- h^2 = 50^2 - 40^2 - M1
- 2500 - 1600 = 900 - M1
- h = 30 (cm) - A1

Total: 3 marks

Question 8

- (a) Mode = 12 - B1
- (b) Range = 20 - 12 = 8 - B1
- (c) Sum = 139; mean = 139/9 = 15.4 - B1

Total: 3 marks

Question 9

- (a) 110 deg is "broadly south-east" (or S 70 E if compass notation accepted) - B1
- (b) 290 - 180 = 110 (deg) - M1
- (b) 110 (deg) - A1

Total: 3 marks

Question 10

- (a) (-3)^2 = 9 ; 2 x 9 + 5 - M1
- (a) y = 23 - A1
- (b) (8 + (-12)) / 2 = -4 / 2 = -2 - B1

Total: 3 marks

Question 11

- (a) Cube root: 5 x 5 x 5 = 125 OR sqrt[3](125) attempted - M1
- (a) 5 (cm) - A1
- (b) 6 x 5^2 = 150 (cm^2) - B1

Total: 3 marks

Question 12

- Multiplier 1.02 seen - M1
- 6500 x 1.02^2 (= 6762.60) seen - M1
- 6500 x 1.02^3 - M1
- £6897.85 (accept £6897.852) - A1

Total: 4 marks

Question 13

- (a) 16 (= 4 x 4) - B1
- (b) Pairs summing to 5: (1,4),(2,3),(3,2),(4,1) - four pairs - M1
- (b) 4 / 16 - M1
- (b) 1/4 - A1

Total: 4 marks

Question 14

- (a) Drop perpendicular; half base = 3 m identified - M1
- (a) h^2 = 5^2 - 3^2 = 25 - 9 = 16 - M1
- (a) h = sqrt(16) = 4 (m) shown - C1
- (b) Area = 0.5 x 6 x 4 - M1
- (b) 12 (m^2) - A1

Total: 5 marks

Question 15

- (a) 30 x 20 x 15 - M1
- (a) 9000 (cm^3) - A1
- (b) 2(30x20) + 2(30x15) + 2(20x15) - M1

- (b) 1200 + 900 + 600 - M1
- (b) 2700 (cm^2) - A1

Total: 5 marks

Question 16

- (a) 6(5) - 1 = 29 - B1
- (b) T_1 = 5; T_10 = 59; sum = 64 - M1
- (b) 64 - A1
- (c) 6n - 1 = 89 -> n = 15 - B1

Total: 4 marks

Question 17

- (a) All 8 points plotted correctly - B2 (B1 if 6+ correct)
- (b) Negative correlation (strong) - B1
- (c) Sensible line of best fit - B1
- (d) Failure rate at 580 (accept 6.5 - 7.5 %) - B1

Total: 5 marks

Question 18

- (a) V = pi x 0.5^2 x 2 (= 3.142 x 0.25 x 2) - M1
- (a) 1.571 (m^3) accept 1.570 - 1.572 - M1
- (a) 1.571 stated to 3 dp - A1
- (b) 1.571 x 1000 - M1
- (b) 1571.0 (kg) - A1

Total: 5 marks

Question 19

- (a) d^2 = 8^2 + 6^2 - M1
- (a) 64 + 36 = 100 - M1
- (a) d = 10 (km) - A1
- (b) 10 km = 1 000 000 cm; / 25 000 - M1
- (b) 40 (cm) - A1

Total: 5 marks

Question 20

- (a) 9 outcomes listed: (3,3)(3,4)(3,7)(4,3)(4,4)(4,7)(7,3)(7,4)(7,7) - B2 (B1 for at least 6 correct)
- (b) Sums identified: 6, 7, 10, 7, 8, 11, 10, 11, 14 - M1
- (b) Even sums identified: 6, 10, 8, 10, 14 (five of nine) - M1
- (b) 5/9 shown clearly - A1

Total: 5 marks

Question 21

- (a) Multiplier 1.05; 2000 x 1.05 = 2100 - M1
- (a) Withdraw £150: 2100 - 150 - M1
- (a) End of year 1: £1950 - A1
- (b) 1950 x 1.05 = 2047.50 - M1

- (b) 2047.50 - 150 - M1
- (b) £1897.50 - A1

Total: 6 marks

Question 22

- (a) 18/32 simplified - M1
- (a) 9/16 - A1
- (b) Independent events identified / multiplication chosen - M1
- (b) (14/32) x (18/32) - M1
- (b) 252/1024 = 63/256 (accept exact value or 0.246 to 3 dp) - A1

Total: 5 marks

Worked Solutions

Foundation Tier - Paper 3 (Calculator) - Set 8 | Detailed explanations

Question 1

10% of £85 = 85 / 10 = £8.50.

Question 2

Divide 23 by 4: 23 / 4 = 5 with remainder 3.

So 23/4 = 5 wholes and 3/4 = 5 3/4.

Question 3

(a) 4837 to 3 sf: keep the first three digits, round using the next: 4837 -> 4840.

(b) 348 to the nearest 100: the tens digit is 4 (< 5), round down -> 300.

Question 4

Make the units the same. 1.5 kg = 1500 g.

Ratio 200 : 1500. Divide both by 100: 2 : 15.

Answer: 2 : 15.

Question 5

(a) Adult row: Total 150, Fiction 85, so Non-fiction = 150 - 85 = 65.

Children total = 240 - 150 = 90. Non-fiction Children = 40, so Fiction Children = 90 - 40 = 50.

Column totals: Fiction 85 + 50 = 135, Non-fiction 65 + 40 = 105. Check: 135 + 105 = 240.

(b) Adult non-fiction = 65. P(adult non-fiction) = 65/240 = 13/48.

Question 6

x - 4 = -7. Add 4 to both sides: x = -3.

Question 7

Pythagoras for the shorter side: h^2 = 50^2 - 40^2 = 2500 - 1600 = 900.

h = sqrt(900) = 30 cm.

Question 8

(a) 12 appears three times (the most). Mode = 12.

(b) Range = largest - smallest = 20 - 12 = 8.

(c) Sum = 14+12+18+17+12+20+15+19+12 = 139. Mean = 139/9 = 15.44..., i.e. 15.4 (1 dp).

Question 9

(a) 110 deg is measured clockwise from north. This is past east (90 deg), so the direction is "broadly south-east".

(b) Reverse bearing: 290 - 180 = 110 deg.

Question 10

(a) Substitute x = -3 into y = 2x^2 + 5.

(-3)^2 = 9. So y = 2 x 9 + 5 = 18 + 5 = 23.

(b) y = (8 + (-12)) / 2 = -4 / 2 = -2.

Question 11

(a) side^3 = 125. Cube root: 5 (since 5 x 5 x 5 = 125). Side = 5 cm.

(b) Surface area = 6 x side^2 = 6 x 25 = 150 cm^2.

Question 12

Multiplier 1.02.

Year 1: 6500 x 1.02 = £6630.00.

Year 2: 6630 x 1.02 = £6762.60.

Year 3: 6762.60 x 1.02 = £6897.852, i.e. £6897.85 to the nearest penny.

Question 13

(a) 4 outcomes per spin, 2 spins, so 4 x 4 = 16 ordered outcomes.

(b) Pairs summing to 5: (1,4), (2,3), (3,2), (4,1) - four pairs.

P(total = 5) = 4 / 16 = 1/4.

Question 14

(a) Drop the perpendicular from the apex to the centre of the base, splitting it into two right-angled triangles with hypotenuse 5 m and shorter side 3 m (half of 6).

h^2 = 5^2 - 3^2 = 25 - 9 = 16. h = sqrt(16) = 4 m. (Shown.)

(b) Area = 0.5 x base x height = 0.5 x 6 x 4 = 12 m^2.

Question 15

(a) V = 30 x 20 x 15 = 9000 cm^3.

(b) Three pairs of identical rectangles.

2(30x20) + 2(30x15) + 2(20x15) = 1200 + 900 + 600 = 2700 cm^2.

Question 16

(a) T_5 = 6(5) - 1 = 29.

(b) T_1 = 5. T_10 = 6(10) - 1 = 59. Sum = 5 + 59 = 64.

(c) $6n - 1 = 89 \to 6n = 90 \to n = 15$. So $T_{15} = 89$.

Question 17

(a) Plot 8 points.

(b) Negative correlation - longer usage time, lower failure rate.

(c) Best fit drawn.

(d) At 580 hr the line gives roughly 6.5 - 7.5 % failure rate.

Question 18

(a) $V = \pi \times r^2 \times h = 3.142 \times 0.25 \times 2 = 1.571\ m^3$.

(b) Mass = 1000 kg per $m^3 \times 1.571\ m^3 = 1571.0$ kg.

Question 19

(a) After the boat sails east then north, S, the boat, and the start point form a right-angled triangle.

$d^2 = 8^2 + 6^2 = 64 + 36 = 100$. $d = 10$ km.

(b) 10 km = 10 000 m = 1 000 000 cm. On map: 1 000 000 / 25 000 = 40 cm.

Question 20

(a) 9 ordered outcomes when drawing twice with replacement from {3, 4, 7}:

(3,3) (3,4) (3,7) (4,3) (4,4) (4,7) (7,3) (7,4) (7,7).

(b) Sums in the same order: 6, 7, 10, 7, 8, 11, 10, 11, 14.

Even sums: 6, 10, 8, 10, 14 - that is 5 outcomes out of 9.

So P(S is even) = 5/9. (Shown.)

Question 21

(a) Add 5% interest: 2000 x 1.05 = £2100.

Withdraw £150: 2100 - 150 = £1950.

(b) Add 5% interest: 1950 x 1.05 = £2047.50.

Withdraw £150: 2047.50 - 150 = £1897.50.

Question 22

(a) P(girl) = 18 / 32. Divide top and bottom by 2: 9/16.

(b) "First boy AND second girl" - replacement implied (any student could be chosen again). The two picks are independent.

P(first boy) = 14/32 = 7/16. P(second girl) = 18/32 = 9/16.

P(both) = (7/16) x (9/16) = 63/256, which is approximately 0.246.